

AMENDMENT NO. 1 DECEMBER 2019

TO

**IS 13364 (PART 2) : 1992 AC GENERATORS DRIVEN
BY RECIPROCATING INTERNAL COMBUSTION
ENGINES — SPECIFICATION**

**PART 2 ALTERNATORS RATED ABOVE 20 kVA
AND UP TO 1 250 kVA**

(Second cover page, Foreword, para 3) — Delete ‘IS 4722 : 1968 Rotating Electrical Machines, issued by Bureau of Indian Standards.’

[Page 1, clause 1.4 (a)] — Delete ‘: 1979’.

(Page 3, clause 4.1.1, line 2) — Delete.

(Page 3, clause 4.1.1) — Insert the following new clause:

‘4.1.1.1 If the machine is to operate at an altitude between 1 000 m and 4 000 m and the maximum coolant temperature is not specified, it shall be assumed that the reduced cooling resulting from altitude is compensated by a reduction of maximum ambient temperature below 40°C and that the limiting total temperature therefore will not exceed 40°C plus the temperature rises as mentioned in Table 7 of IS 15999 (Part 1). The specified altitude and assumed maximum ambient temperature shall be marked on the rating plate.

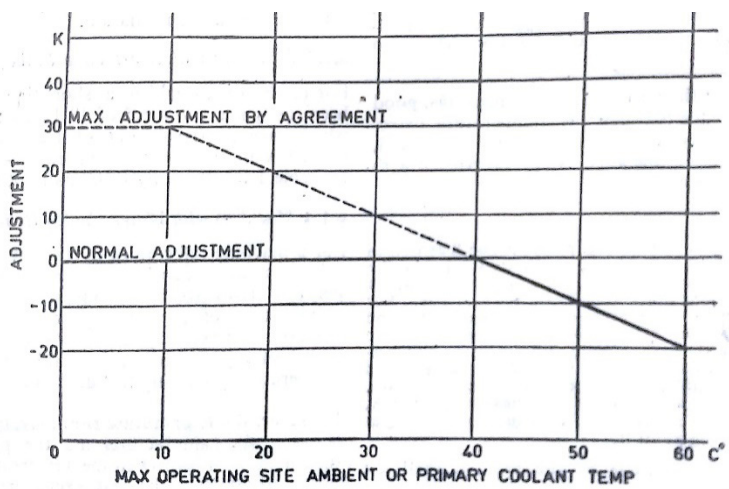
If the machine is to operate at an altitude exceeding 4 000 m the maximum limits of temperature rise shall be agreed between the manufacturer and the purchaser.’

(Page 3, clause 4.1.2, line 2) — Delete.

(Page 3, clause 4.1.2) — Insert the following new clause:

‘4.1.2.1 if the specified or resulting maximum coolant temperature exceeds 60°C or is less than 0°C, the limits of temperature rise shall be agreed between the manufacturer and the purchaser.

If the specified or resulting maximum coolant temperature is between 40°C and 60°C, the limits of temperature rise given in Table 7 of IS 15999 (Part 1) shall be reduced by the amount by which the coolant temperature exceeds 40°C. This is illustrated as below:



(Page 3, clause 5) — Substitute ‘IS/ IEC 60034 (Part 5)’ for ‘IS 4691 : 1985’.

(Page 3, clause 6) — Delete ‘: 1971’.

(Page 4, clause 8.8, title) — Substitute ‘IRREGULARITIES OF WAVEFORM’ for ‘IRREGULARITIES OR WAVEFORM’.

(Page 4, clause 8.3) — Substitute the following for the existing:

‘8.3 Telephone Harmonic Factor

The requirements of this clause apply only to synchronous machines of 300 kVA or above, intended for connection to power networks operating at rated frequencies of 16.67 Hz to 100 Hz inclusive, with a view to minimizing interference between power lines and adjacent circuits.

8.3.1 Requirements

When tested on open circuit and at rated speed and voltage, the telephone harmonic factor (THF) of the line-to-line terminal voltage as measured according to 8.3.2 shall not exceed the following value:

<i>Rated Output</i>	<i>THF Percent</i>
300 kVA < P ≤ 1 000 kVA	5
1 000 kVA < P ≤ 1 250 kVA	3

NOTES

1 Limiting values of individual harmonics are not specified as it is considered that machines which meet the above requirements shall be operationally satisfactory,

2 Where the synchronous machine is to be connected to the system in an unusual manner (for example, where the star point of machine is connected to earth and the machine is not linked to the system via a transformer), the waveform requirements should be agreed between the manufacturer and the purchaser.

8.3.2 Tests conformity with 8.3.1, shall be verified by type tests on a.c. generators. The range of frequencies measured shall cover all harmonics from rated frequency up to 5 000 Hz. Either the THF may be measured directly by means of a meter and associated network specially designed for the purpose or each individual harmonic shall be measured and from the measured values the THF shall be computed using the following formula:

$$THF \text{ (percent)} = \frac{100}{U} \sqrt{E_1^2 \gamma_1^2 + E_2^2 \gamma_2^2 + E_3^2 \gamma_3^2 + \dots E_n^2 \gamma_n^2}$$

where

E_n = rms value of nth harmonic line-to-line terminal voltage,

U = rms value of line-to-line terminal voltage of machine, and

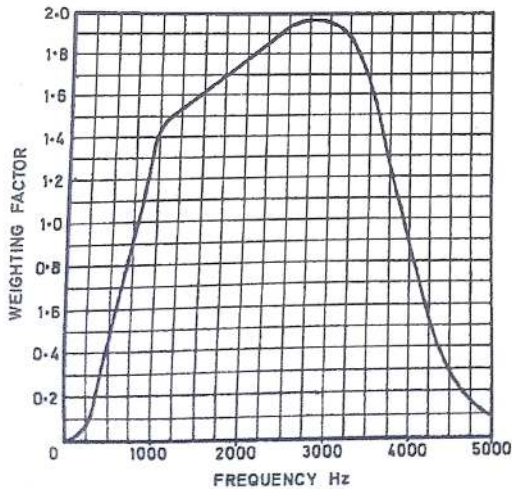
γ_n = weighting factor for frequency corresponding to nth harmonic.

Numerical values of the weighting factor for different frequencies shall be obtained from table as shown below:

<i>Weighting Factors</i>					
Sl No.	Frequency (Hz)	Weighting Factor	Sl No.	Frequency (Hz)	Weighting Factor
(1)	(2)	(3)	(4)	(5)	(6)
1)	16.66	0.000 001 17	28)	1 350	1.57
2)	50	0.000 044 4	29)	1 400	1.58
3)	100	0.001 12	30)	1 450	1.60
4)	150	0.006 65	31)	1 500	1.61
5)	200	0.022 3	32)	1 550	1.63
6)	250	0.055 6	33)	1 600	1.65
7)	300	0.111	34)	1 650	1.66
8)	350	0.165	35)	1 700	1.68
9)	400	0.242	36)	1 750	1.70
10)	450	0.327	37)	1 800	1.71
11)	500	0.414	38)	1 850	1.72
12)	550	0.505	39)	1 900	1.74
13)	600	0.585	40)	1 950	1.75
14)	650	0.691	41)	2 000	1.77
15)	700	0.790	42)	2 050	1.79
16)	750	0.895	43)	2 100	1.81
17)	800	1.000	44)	2 150	1.82
18)	850	1.10	45)	2 200	1.84
19)	900	1.21	46)	2 250	1.86
20)	950	1.32	47)	2 300	1.87
21)	1 000	1.40	48)	2 350	1.89
22)	1 050	1.46	49)	2 400	1.90
23)	1 100	1.47	50)	—	—
24)	1 150	1.49	51)	2 450	1.91
25)	1 200	1.50	52)	2 500	1.93
26)	1 250	1.53	53)	2 550	1.93
27)	1 300	1.55	54)	2 600	1.94

Weighting Factors					
Sl No.	Frequency (Hz)	Weighting Factor	Sl No.	Frequency (Hz)	Weighting Factor
(1)	(2)	(3)	(4)	(5)	(6)
55)	2 650	1.95	69)	3 700	1.35
56)	2 700	1.96	70)	3 800	1.19
57)	2 750	1.96	71)	3 900	1.04
58)	2 800	1.97	72)	4 000	0.890
59)	2 850	1.97	73)	4 100	0.740
60)	2 900	1.97	74)	4 200	0.610
61)	2 950	1.97	75)	4 300	0.496
62)	3 000	1.97	76)	4 400	0.960
63)	3 100	1.94	77)	4 500	0.316
64)	3 200	1.89	78)	4 600	0.232
65)	3 300	1.83	79)	4 700	0.199
66)	3 400	1.75	80)	4 800	0.138
67)	3 500	1.65	81)	4 900	0.125
68)	3 600	1.51	82)	5 000	0.100

The weighting curve as follows may be used as an aid to interpolation.



WEIGHTING CURVE'

(Page 5, clause **9.1**, Para 2, line 2) — Substitute ‘IS 15999 (Part 1)’ for ‘IS 4722 : 1992’.

(Page 5, clause **10**) — Delete ‘: 1987’.

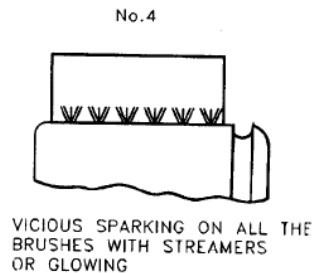
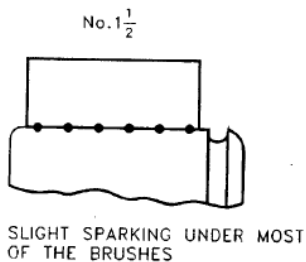
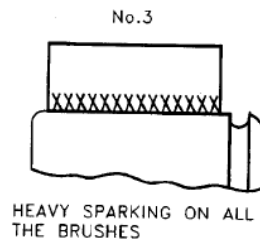
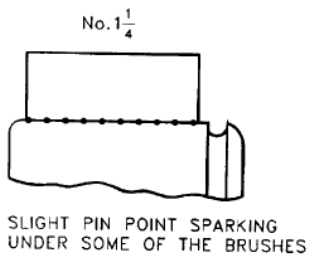
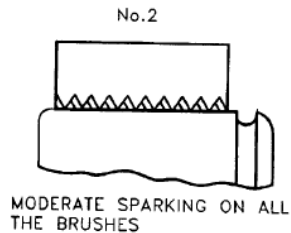
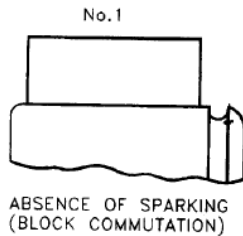
(Page 5, clause **11**) — Substitute ‘8 of IS 15999 (Part 1)’ for ‘IS 4722 : 1992’.

(Page 5, clause **14.1**, line 3) — Substitute ‘commutator’ for ‘computator’.

(Page 5, clause **14.3**) — Substitute the following for the existing clause:

14.3 Grades

‘14.3.1 The following are the grades of sparking:



14.3.1.1 No. 1

No blackening of commutator and burning of brushes.

14.3.1.2 No. 1 $\frac{1}{2}$ and No. 1 $\frac{1}{4}$

Traces of black spots on commutator which can be removed by petrol. Traces of burning on brushes.

14.3.1.3 No. 2

Permitted only for short time overloads. Trace of black spots on commutator which cannot be removed by petrol. Traces of burns on brushes.

14.3.1.4 No. 3

Permitted only for direct on load starting and for the starting period. Easily spottable blackening of commutator which cannot be removed by petrol. Burning and slight damaging of brushes.

14.3.1.5 No. 4

Too much blackening of commutator which cannot be removed by petrol. Damaging of brushes.

14.3.2 Grades No. 1, 1₂¹, 1₄¹ and 2 shall be used in specifying the degree of sparking. Grades No. 1₂¹ and 1₄¹ shall be rectified/maintained weekly and Grade No. 2 shall be rectified/maintained as soon as sparking is noticed.

NOTE — Grades No. 3 and 4 are not recommended for operation.'

(Page 6, clause 17) — Substitute 'IS/IEC 60034 (Part 8)' for 'IS 4728 : 1975'.

(Page 7, clause 20.2, line 2) — Substitute 'IS 15999 (Part 4)' for 'IS 7306 : 1974'.

(Page 7, clause 20.3) — Substitute the following for the existing:

‘20.3 Synchronous Generators

20.3.1 Type Tests

The following shall constitute the type tests:

- a) Measurement of resistance (*see* 8 of IS 7132);
- b) Phase sequence test (for 3-phase generators only) (*see* 11 of IS 7132);
- c) Regulation test (*see* 15);
- d) Measurement of open circuit characteristic only (*see* 21 of IS 7132);
NOTE — The exciter field current and voltage shall be measured for brushless alternator.
- e) Measurement of short-circuit characteristic only (*see* 21 of IS 7132);
NOTE — The exciter field current and voltage shall be measured for brushless alternator.
- f) Efficiency test (*see* 15);
- g) Temperature rise test (*see* 11);
- h) Overspeed test (*see* 23);

- j) Insulation resistance test (both before and after high voltage test) (*see 22*);
- k) High voltage test (*see 21*);
- m) Determination of deviation of voltage wave-form from sinusoidal (subject to agreement between the manufacturer and the user) (*see 8*);
- n) Momentary overload test (*see 12.1*); and
- p) Test for vibration severity (*see 13*) (under consideration).

20.3.2 Routine Tests

The following shall constitute the routine tests:

- a) Measurement of resistance (*see 8 of IS 7132*);
- b) Insulation resistance test (*see 22*);
- c) Phase sequence test (for 3-phase generators only) (*see 11 of IS 7132*);
- d) Regulation test (*see 15*);
- e) Measurement of open circuit characteristic (*see 21 of IS 7132*);
- f) Measurement of short-circuit characteristic (*see 21 of IS 7132*); and
- g) High voltage test (*see 21*).’

(*Page 7, clause 21*) — Substitute ‘9.2 of IS 15999 (Part 1)’ for ‘IS 4722 : 1992’.

(*Page 7, clause 22*) — Substitute ‘IS/IEC 60034-27-4’ for ‘IS 4722 : 1992’.

(*Page 7, clause 24, line 6*) — Substitute ‘IS 15999 (Part 1)’ for ‘IS 4722 : 1992’.

(*Page 8, clause 29*) — Substitute the following for the existing:

IS 900 : 1992	Code of practice for installation and maintenance of induction motors
IS 1076 (Part 1) : 1985	Preferred Numbers: Part 1 Series of preferred numbers (<i>second revision</i>)
IS 12360 : 1988/ IEC 60038	Voltage bands for electrical installations including preferred voltages and frequency

- IS 13584 : 1993 Brush materials for electrical machinery
- IS 13586 : 1993 Definitions and nomenclature for carbon brushes, brush holders, commutators and slip rings for rotating electrical machinery

(Page 8, clause **30**, line 1) — Delete ‘ : 1974’.

(Page 9, Annex A) — Substitute the following for the existing:

ANNEX A

(Clause 2)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
1231 : 1974	Dimensions of Three-phase Foot-mounted Induction Motors (<i>Third Revision</i>)	7132 : 1973	Guide for testing synchronous machines
3043 : 2018	Code of Practice for Earthing (<i>Second Revision</i>)	10242 (Part 3/Sec 1) : 1983	Electrical installations in ships: Part 3 Equipment, Section 1 Generators and motors
4889 : 1968	Methods of Determination of Efficiency of Rotating Electrical Machines	12075 : 2008	Mechanical Vibration of Rotating Electrical Machines with Shaft Heights 56 mm and Higher — Measurement, Evaluation and Limits of Vibration Severity (<i>First Revision</i>)
5422 : 1996	Turbine Type Generators (<i>First Revision</i>)	15999 (Part 1) : 2016	Rotating Electrical Machines: Part 1 Rating and Performance (<i>First Revision</i>)’
6362 : 1995	Designation of Methods of Cooling of Rotating Electrical Machines (<i>First Revision</i>)		

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
15999 (Part 4) : 2017	Rotating electrical machines: Part 4 Methods for determining synchronous machine quantities from tests	60034 (Part 8) : 2002	Rotating electrical machines: Part 8 Terminal markings and direction of rotation (<i>Second Revision</i>)
60034 (Part 5) : 2000	Rotating electrical machines: Part 5 Degrees of protection provided by the integral design of rotating electrical machines (IP CODE) — Classification (<i>Second Revision</i>)	IS/IEC 60034-27-4 : 2018	Rotating electrical machines: Part 27-4: Measurement of insulation resistance and polarization index of winding insulation of rotating electrical machines (<i>Under print</i>)